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LETTER TO THE EDITOR

Vibronic shifts for LiH in X and A states

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Abstract. The energy shift of the vibrational levels of the X and A states due to the vibronic interactions within the manifold of $^1\Sigma$ states is accurately determined using a diabatic representation. These vibronic shifts which measure the breakdown of the Born–Oppenheimer approach are compared with the usual estimation where only the adiabatic correction is taken into account. The adiabatic corrections lead to a systematic overestimation of the full vibronic shift for the ground state and to an underestimation for the A one. These results have been related to the interstate dynamical couplings and rationalized using perturbative arguments. Their validity is expected to be very general.

Vibronic effects are known to be important when potential energy surfaces are close together in energy and particularly when they become degenerate for some configuration like in a conical intersection. Such situations are not rare events and Jahn–Teller or Renner–Teller effects are rather common in molecular physics. The interplay between the nuclear and electronic motions plays an essential role and vibronic models have been developed [1] for its description.

Indeed, avoided crossings are the rule rather than the exception and imprint most of the usual Born–Oppenheimer energy surfaces. It could thus be expected that vibronic effects play some role; however, as the crossing becomes largely avoided and as the nuclear masses increase, these effects which break the Born–Oppenheimer approximation rapidly decrease. They are, in fact, considered to be minor effects relevant mainly for high-accuracy spectroscopy and are often ignored. Comparison with the experimentally determined energy levels is then performed within the BO separation as in the widely used Rydberg–Klein–Rees (RKR) method.

Since the pioneering study [2] by Kolos and Wolniewicz on the H_2 molecule, we know that the potential energy curves can become more accurate when the adiabatic corrections are taken into account. However, this correction rests upon the general BO scheme and each adiabatic potential is considered independently of the others. Only for the evaluation of the correction is the mixing considered and the comparison with experimental results relies on the usual uncoupled inversion methods.

To go beyond this, the coupled equations now involving the whole matrix of the nuclear kinetic operator in the electronic representation (and not only its diagonal elements) should be solved, i.e. the vibronic approach. In the present contribution, we compare the usual BO results with those obtained within a close-coupling vibronic approach involving the

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eight $^1\Sigma$ lowest diabatic electronic states for LiH. We calculate vibronic shifts for the vibrational levels of the X and A states and compare them with those arising from the adiabatic correction.

The two lowest adiabatic states of LiH result from an avoided crossing between the ionic Li^+H^- and covalent Li(2s)H states. This crossing is largely avoided and similar ones can be found for a broad class of molecules, in such a way that no particular vibronic effects are expected here. We will see, however, that their contribution is significant and can even be larger than the one due to adiabatic corrections. Therefore, when high accuracy is required, the usual inversion methods, which ignore these effects, are no longer applicable and more general inversion procedures involving several coupled electronic states have to be used.

First the various theoretical approaches are presented, and then their application to LiH is presented and discussed.

For completeness and in order to introduce the notations, we recall the theoretical background for the treatment of vibronically coupled electronic states. As usual we follow a close-coupling approach where only n electronic states are considered. For each interatomic distance R, n adiabatic states ($\psi_i, i = 1, n$) which diagonalize the BO electronic Hamiltonian span the electronic representation. General electronic basis sets ($\phi_i, i = 1, n$) can be obtained from some unitary transformation at each R value,

$$\phi_i = \sum_j U_{ij}(R)\psi_j \tag{1}$$

$$(U^\dagger U = 1)\,. \tag{2}$$

The vibrational basis set for R is given by N orthonormal functions ($\chi_v, v = 1, N$) and will be specified later on. The full representation results from the tensor product of electronic and vibrational spaces $|\Psi_m\rangle = |\phi_i\rangle \otimes |\chi_v\rangle$. The total Hamiltonian is

$$H = \sum_{m,n} |\Psi_m\rangle\langle\Psi_m|(T + V)|\Psi_n\rangle\langle\Psi_n| \tag{3}$$

where $T = -(1/2\mu)\partial^2/\partial R^2$ is the nuclear kinetic energy operator in the non-rotating diatomic approximation and V is the electronic Hamiltonian including the nuclear repulsion.

In the adiabatic representation (ψ) V is diagonal

$$\langle\psi_i|V|\psi_j\rangle = E_i(R)\delta_{ij} \tag{4}$$

and to get the vibronic levels, one has to diagonalize the following matrix:

$$H_{iv,jv'} = \langle\chi_v|\left[\delta_{ij}\left(-\frac{1}{2\mu}\frac{\partial^2}{\partial R^2} + E_i(R)\right) - \frac{1}{2\mu}\left(h_{ij}(R) + 2f_{ij}(R)\frac{\partial}{\partial R}\right)\right]|\chi_{v'}\rangle \tag{5}$$

where

$$f_{ij} = \left\langle\psi_i\left|\frac{\partial}{\partial R}\psi_j\right.\right\rangle \tag{6}$$

and

$$h_{ij} = \left\langle\psi_i\left|\frac{\partial^2}{\partial R^2}\psi_j\right.\right\rangle. \tag{7}$$

Since f is antisymmetric, as follows from normalization and orthogonality of the ψ functions, only h presents nonvanishing diagonal elements. They can eventually be included in the diagonal elements of the electronic Hamiltonian. When the adiabatic electronic representation is used, these diagonal elements contribute to the adiabatic correction. Other terms, such as mass polarization or finite mass corrections, also contribute to the adiabatic

correction. However, here we will consider only that part due to the variation with R of the coefficients U and therefore h_{ii} vanishes for large R as U approaches the unit matrix. From the definition of f, inserting twice the closure relation

$$1_{nn} = \sum_{i=1}^{n} |\psi_i\rangle\langle\psi_i| \tag{8}$$

differentiating with respect to R and using the antisymmetry of the f matrix, we get

$$h_{ij} = (f^2)_{ij} + \frac{\partial f_{ij}}{\partial R}\,. \tag{9}$$

On the other hand, a diabatic scheme (φ) is that in which T becomes diagonal

$$\langle\varphi_i|T|\varphi_j\rangle = -\frac{1}{2\mu}\frac{\partial^2}{\partial R^2}\delta_{ij} \tag{10}$$

but V presents diagonal as well as off-diagonal terms and the matrix to be diagonalized is now

$$H_{iv,jv'} = \langle\chi_v|\left[\delta_{ij}\left(-\frac{1}{2\mu}\frac{\partial^2}{\partial R^2}\right) + \langle\varphi_i|V|\varphi_j\rangle\right]|\chi_{v'}\rangle\,. \tag{11}$$

In this work we assume known diabatic states (φ); they have been determined directly during the *ab initio* calculation [3, 4] according to a procedure [5] which combines the variational effective Hamiltonian [6] and effective metric approaches. Using the diabatic representation, the matrix elements $f_{ij}(R)$ and $h_{ij}(R)$ can be easily derived. To evaluate f, a convenient way is to use an Helmann–Feynman expression

$$f_{ij}(R) = \frac{1}{E_i(R) - E_j(R)}[U^{\dagger}(R)A(R)U(R)]_{ij} \tag{12}$$

where $A(R)$ is the matrix representation of $\partial V(R)/\partial R$ in the diabatic basis set. There is no difficulty in the evaluation of the derivative of the electronic couplings in the diabatic representation since they behave as smooth functions of R. For LiH, this estimation [7] of the radial coupling compares favourably to the direct *ab initio* calculation [8]. In addition, the physical origin of the peaks in the radial coupling is clarified by a clear correlation with the crossings in the diabatic curves. This study [7] has confirmed the known neutral–ionic crossings and revealed unexpected neutral–neutral ones at short distances.

Both terms f_{ij} and h_{ij} vanish in the diabatic representation and may diverge in the adiabatic one as can be seen from the Hellman–Feyman expression when the energy difference in the denominator is small. Of the diagonal terms only $h_{ii} = (f^2)_{ii}$ remains, which is negative because of the antisymmetry of the f matrix.

In the Born–Oppenheimer approximation, the adiabatic representation is used and both f and h are neglected. The vibronic matrix is then block diagonal and the vibrational levels can be determined for each adiabatic potential separately. This result will be labelled as E_v^a. Still neglecting the interblock couplings due to f and h, the diagonal matrix elements $-(1/2\mu)h_{ii}(R)$ can be added to the adiabatic curves $E_i(R)$ and each potential treated separately again. This result will now be labelled E_v^{ac}. Finally, the whole vibronic matrix in the diabatic representation can be diagonalized leading to the exact results denoted by E_v.

We have tested various vibrational basis sets. The most reliable results have been obtained when using the eigenfunctions corresponding to the diabatic ionic curve ($V^I(R)$), i.e. those satisfying the equation

$$\left[-\frac{1}{2\mu}\frac{\partial^2}{\partial R^2} + V^I(R)\right]\chi_v(R) = E_v^I\chi_v(R)\,. \tag{13}$$

They have been accurately determined by means of a Numerov algorithm over a grid of about 20 000 points. The reason for this choice is that its spans the whole energy range of interest and leads to bound states which span the whole R range of interest. Using the diabatic representation it is not necessary to compute the nuclear kinetic operator again since we have

$$\langle \varphi_i \chi_v | T | \varphi_j \chi_{v'} \rangle = \delta_{ij} (E_v^I \delta_{vv'} + \langle \chi_v | V_{ii}(R) - V^I(R) | \chi_{v'} \rangle) . \tag{14}$$

In this way, only potential-like integrals have to be evaluated for the full vibronic approach. The matrix to be diagonalized is of dimension $n \times N$. Here, for LiH, $n = 8$ and we have varied N between 100 and 120. The resulting matrix is moderate in size and the accuracy of the calculation can be estimated from the variation of N. The bound and quasi-bound functions are practically unaffected by this variation, since their energy change for the X and A states is only about 10^{-5} cm^{-1}, while the energy of the discretized continuum states varies much more. Thus, the variation of N allows for an easy identification of bound and quasi-bound states as is usual in stabilization methods [9]. Large DVR grids have also been tested (with N about 300). They lead to larger vibronic matrices and, moreover, the bound and quasi-bound states are stabilized only to 10^{-2}–10^{-3} cm^{-1}, an accuracy two orders of magnitude worse than the approach we have adopted.

Adiabatic and diabatic *ab initio* curves are reported in figures 1 and 2, respectively. The large electronic couplings make the neutral–ionic crossing around 6 au largely avoided. It leads to a peak in the radial coupling for the lowest adiabatic states X and A (figure 6

Table 1. Vibrational levels and shifts for the X state, energies are in cm^{-1} with respect to the X asymptote. v, vibrational level; E^a, adiabatic Born–Oppenheimer result; E, exact vibronic result; E^{ac}, as in E^a now including adiabatic corrections, vibronic shift ($E - E^a$), adiabatic correction shift ($E^{ac} - E^a$).

v	E^a	E	E^{ac}	$E - E^a$	$E^{ac} - E^a$
0	−19 460.081	−19 458.804	−19 458.725	1.278	1.356
1	−18 116.215	−18 114.752	−18 114.562	1.463	1.653
2	−16 816.295	−16 814.767	−16 814.459	1.528	1.836
3	−15 559.529	−15 557.971	−15 557.568	1.558	1.961
4	−14 344.598	−14 343.047	−14 342.544	1.552	2.054
5	−13 170.596	−13 169.083	−13 168.462	1.513	2.134
6	−12 037.033	−12 035.566	−12 034.812	1.467	2.220
7	−10 943.596	−10 942.159	−10 941.265	1.437	2.331
8	−9 890.127	−9 888.687	−9 887.650	1.440	2.477
9	−8 876.716	−8 875.233	−8 874.054	1.483	2.662
10	−7 903.634	−7 902.074	−7 900.750	1.560	2.884
11	−6 971.293	−6 969.631	−6 968.158	1.662	3.135
12	−6 080.281	−6 078.497	−6 076.867	1.784	3.413
13	−5 231.646	−5 229.727	−5 227.931	1.919	3.715
14	−4 427.020	−4 424.961	−4 422.991	2.059	4.029
15	−3 668.552	−3 666.360	−3 664.209	2.192	4.343
16	−2 959.386	−2 957.078	−2 954.751	2.308	4.635
17	−2 304.152	−2 301.769	−2 299.293	2.383	4.859
18	−1 709.009	−1 706.628	−1 704.064	2.380	4.945
19	−1 182.203	−1 179.938	−1 177.398	2.265	4.806
20	−734.635	−732.629	−730.287	2.006	4.348
21	−380.024	−378.433	−376.519	1.591	3.505
22	−134.558	−133.528	−132.293	1.029	2.265
23	−14.065	−13.735	−13.362	0.331	0.703

Table 2. Vibrational levels and shifts for the A state, energies in cm^{-1} with respect to the X asymptote. v, vibrational level; E^a, adiabatic Born–Oppenheimer result; E, exact vibronic result; E^{ac}, as in E^a now including adiabatic corrections, vibronic shift ($E - E^a$), adiabatic correction shift ($E^{ac} - E^a$).

v	E^a	E	E^{ac}	$E - E^a$	$E^{ac} - E^a$
0	6 358.227	6 361.372	6 361.346	3.144	3.118
1	6 637.557	6 640.891	6 640.808	3.334	3.251
2	6 948.937	6 952.551	6 952.376	3.614	3.439
3	7 282.841	7 286.694	7 286.396	3.853	3.555
4	7 633.969	7 638.031	7 637.586	4.062	3.617
5	7 998.020	8 002.252	8 001.645	4.231	3.624
6	8 371.748	8 376.111	8 375.335	4.363	3.587
7	8 752.649	8 757.115	8 756.170	4.466	3.521
8	9 138.390	9 142.933	9 141.826	4.543	3.436
9	9 526.951	9 531.552	9 530.292	4.600	3.341
10	9 916.649	9 921.292	9 919.894	4.643	3.245
11	10 305.877	10 310.550	10 309.030	4.674	3.153
12	10 693.210	10 697.905	10 696.280	4.695	3.070
13	11 077.309	11 082.018	11 080.308	4.709	2.999
14	11 456.882	11 461.599	11 459.827	4.717	2.945
15	11 830.638	11 835.356	11 833.547	4.718	2.909
16	12 197.268	12 201.983	12 200.165	4.715	2.897
17	12 555.373	12 560.081	12 558.283	4.708	2.909
18	12 903.354	12 908.047	12 906.302	4.693	2.949
19	13 239.274	13 243.938	13 242.286	4.664	3.012
20	13 560.735	13 565.344	13 563.825	4.609	3.090
21	13 864.668	13 869.169	13 867.827	4.501	3.160
22	14 146.858	14 151.157	14 150.033	4.299	3.175
23	14 401.257	14 405.188	14 404.312	3.931	3.055
24	14 619.062	14 622.373	14 621.753	3.310	2.690
25	14 787.092	14 789.432	14 789.053	2.340	1.961
26	14 886.452	14 887.429	14 887.277	0.976	0.824

of [7]). The radial coupling A–C (figure 7 of [7]) presents two peaks assigned [7] to a neutral–neutral crossing at short distances and a neutral–ionic one at large distances. These crossings provide a contribution to the adiabatic corrections h_{ii}. This contribution is positive by construction (equation (9)) and vanishes for large R. Its order of magnitude is the wavelength. The full adiabatic corrections have been determined for LiH by various authors [10–12]. They present a large atomic contribution which is not taken into account here. However, the variations with R of the adiabatic corrections are mainly due to the mixing of electronic states. The present estimate derives from the diabatic-to-adiabatic representation switching. Although it only accounts for this mixing it leads to a correct magnitude for the variations of the adiabatic corrections.

In order to test the sensitivity of the results to the details of the potentials a correction to the ionic diabatic curve has been used. This correction accounts for the hydrogen electroaffinity error due to the basis set limitation and can be easily cast in the diabatic representation [3]. However, for the vibronic shifts, only minor changes with respect to the main conclusions have been observed and we report only the results including the hydrogen electroaffinity correction.

The bound state vibrational levels resulting from the various approaches are reported in table 1 for the ground state and in table 2 for the A one. These approaches are: (i) the

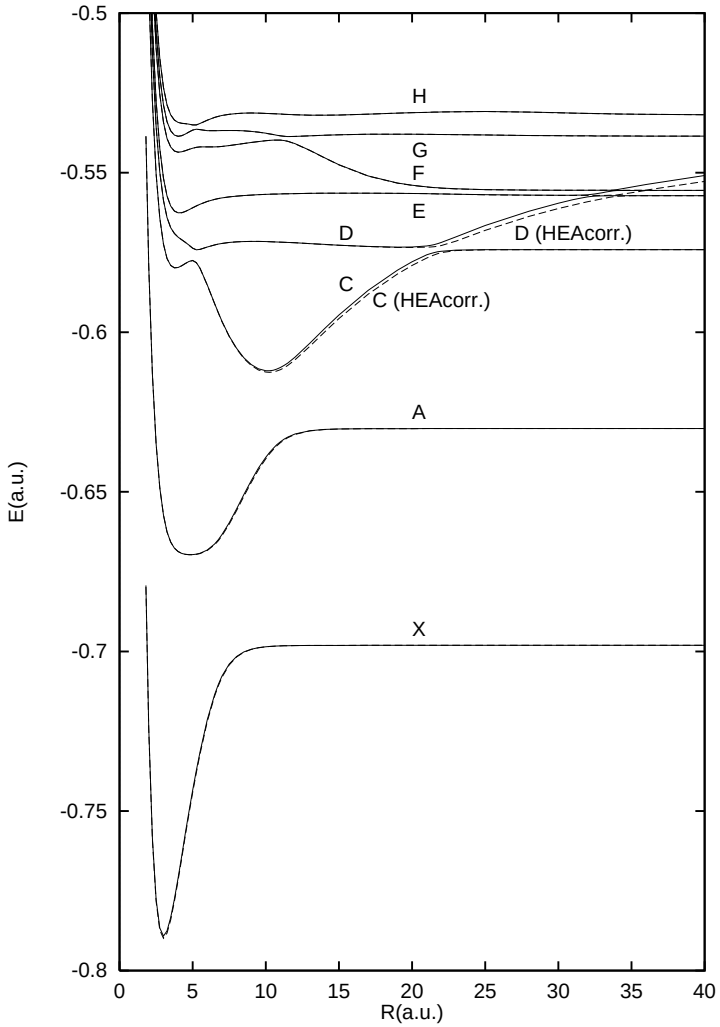


Figure 1. Potential energy curves for LiH $^1\Sigma$ states in the adiabatic representation with and without the correction for the electroaffinity of hydrogen.

usual adiabatic BO; (ii) as (i) but including the adiabatic correction and (iii) a full vibronic matrix. In the vibronic approach the reported levels for the A state correspond to resonances which are stable to within 10^{-5} cm^{-1}. This remarkable stability shows that the coupling with the X continuum is very weak here and does not contribute significantly to the energy shift. The energy shift for approaches (ii) and (iii) with respect to the usual (i) approach is also reported in tables 1 and 2. For the ground state vibrational levels the vibronic shifts are comparable to those due to the adiabatic correction, but are systematically smaller. The adiabatic correction increases the adiabatic energy, it is thus not surprising that all the shifts are positive. The following inequality holds for all the vibrational levels of the X state:

$$E_v^a < E_v < E_v^{ac} . \tag{15}$$

For the A state also the shifts have a positive sign and the adiabatic correction accounts to a large extent for the vibronic shift. However, for all the vibrational levels, the vibronic

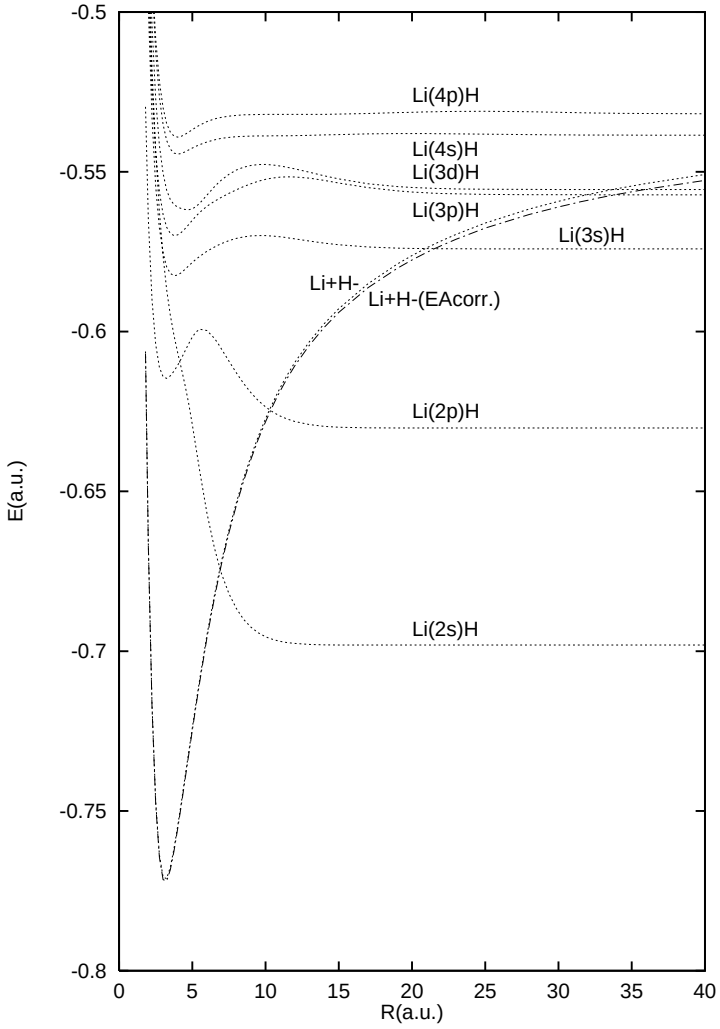


Figure 2. Potential energy curves for LiH $^1\Sigma$ states in the diabatic representation with and without the correction for the electroaffinity of hydrogen.

shift is now larger than the adiabatic correction shift and we have

$$E_v^a < E_v^{ac} < E_v \,. \tag{16}$$

Taking into account the hydrogen electroaffinity error, the ionic curve is shifted to lower energies and the well of the ground state is deeper, while that of the A state is made wider. The absolute position of the vibrational levels is clearly affected in all approaches but the main conclusions for the vibronic shifts remain. The vibronic shifts for the ground and A states are comparable with the adiabatic correction shifts, being smaller for the ground state and larger for the A one. In both cases the vibronic shifts present a regular shape as a function of v and the energy differences between consecutive vibrational levels are only slightly affected. The residual error with respect to the experimental results is largely due to inaccuracies in the *ab initio* potential calculation, but a more detailed comparison is beyond the scope of the present work.

The different behaviour for the X and A states should be related to the interblock non-Born–Oppenheimer couplings f_{ij} and h_{ij} which are ignored in (ii) where only h_{ii} acts. For the ground state, they all act in the same direction, against the adiabatic correction. In a perturbative approach it can be easily seen that, for all interatomic distances, all denominators have a negative sign and thus lead to a contribution which decreases the energies E_v^{ac}. The vibrational levels of the ground state which are the only ones really bound, are stabilized when the dynamical couplings are fully taken into account with respect to the adiabatic correction where only their diagonal contribution remains. For the A state and also for the ones higher in energy, there is a large compensation between the interblock dynamical contributions. The diagonal intrablock one dominates. The final sign of the interblock contributions cannot be easily anticipated. We observe here that for the A state, the contribution due to the coupling with the ground state dominates since we have $E_v^{ac} < E_v$.

In conclusion, using a stabilization method we have accurately determined the vibronic shifts for the vibrational levels of LiH in the ground and A states. They introduce significant displacements, of the order of some wavenumbers, and the adiabatic corrections account, to a large extent, for the full vibronic shifts. We observe remarkable inequalities, which are valid for all the vibrational states and which differ for the ground and A states. The vibronic shift is smaller than the adiabatic correction shift for the vibrational levels of the ground state, while it is larger for the vibrational levels of the A state. We notice that only the vibrational levels of the ground state are bound, those of the A state and also the states higher in energy corresponding to resonances. The different behaviour for the bound states and the resonances has been rationalized using perturbative arguments and related to the interstate dynamical couplings. It is interesting to point out that these interstate dynamical couplings are also responsible for the finite nonradiative lifetime of the resonances. These observations are not specific for the LiH diatomic molecule but should be valid for all molecular systems.

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